Section 6: Solution Statement and Defining Subsystems

6.1. Final Problem Statement

As increasingly concentrated biologic formulations for intravitreal injections become more prevalent, the solutions also become more viscous. Due to the viscosity, injection forces during an administered dosage using standard off-the-shelf syringes increase, resulting in increased injection difficulty. There is a need for a device that can reliably integrate with syringes to reduce injection forces and ensure optimal injection form.

6.2. Solution Statement

Iteration 1: Oct 28, 2026

We will develop a reusable force-amplification device that clips onto a standard 1 mL Luer-lock syringe with a 30G needle. Within the device, a Scott Russell linkage converts a one-handed press by the thumb on the driving lever arm into plunger force. A fixed frame holds the syringe steady, and a linear guide rail keeps the slider aligned and limits its travel to the prescribed dose. Proof-of-concept calculations show force amplification around 37.85°. We want to target a hand force of 10 N[1] to 50 N[2] for the actual device. A tapered needle is fitted ahead of the 30G lumen, and the device can be sterilized for reuse. Our approach is innovative because it is purely mechanical and needs no power. Unlike autoinjectors and electronic systems, it works with off-the-shelf syringes while keeping the clinician in direct, tactile control of the injection.

6.3. **Drawing(s) of Prototype Solution**

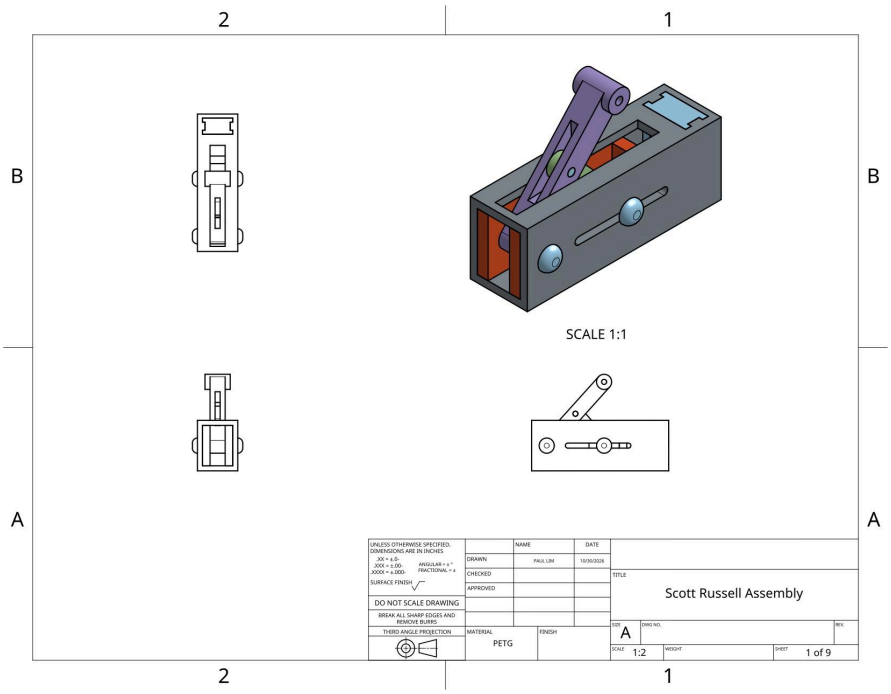


Figure 1. Fully assembled prototype, dimensions not labeled due to complexity and space constraints. See Figures 2-9 for dimensions

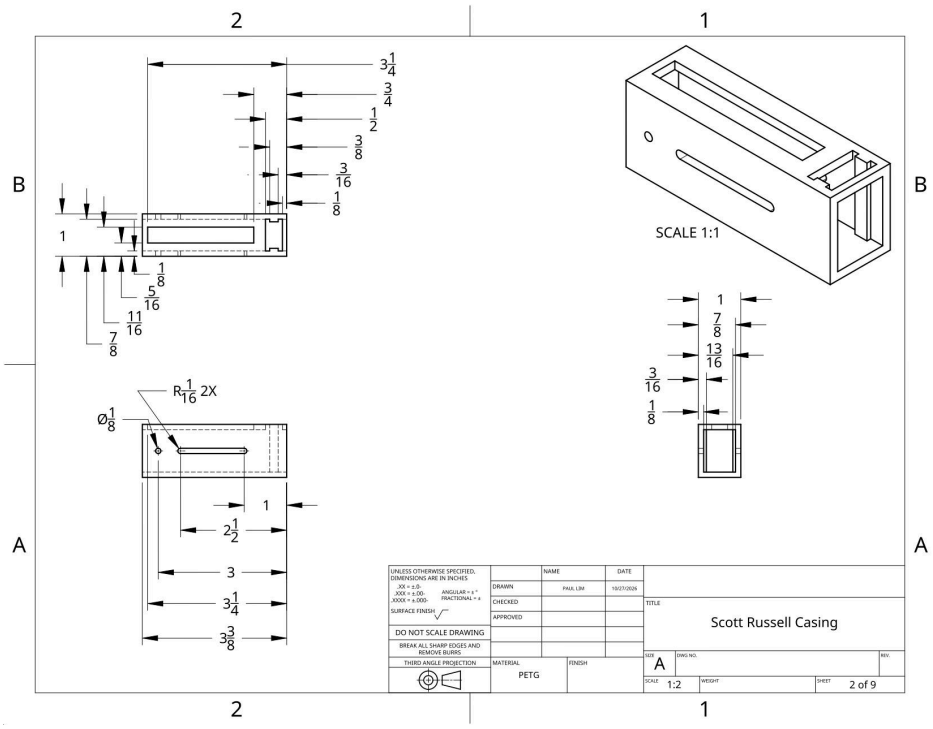


Figure 2. Dimensions and sketches of prototype's casing

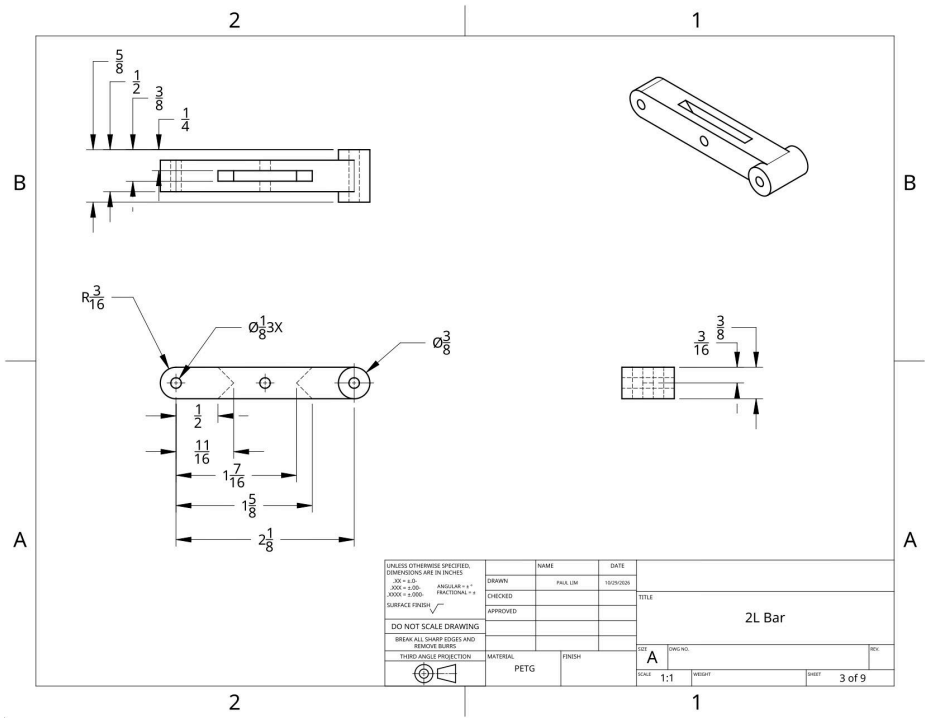


Figure 3. Dimensions and sketches of prototype bar of double length, a main component of the scott russell linkage

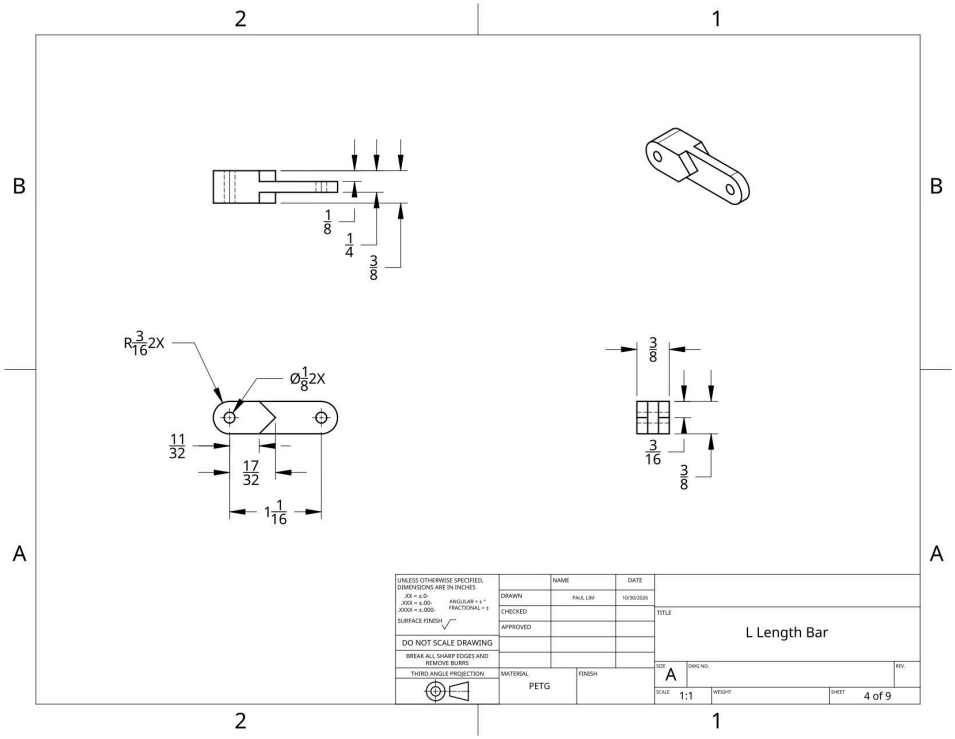


Figure 4. Dimensions and sketches of prototype bar of single length, another main component of the scott russell linkage

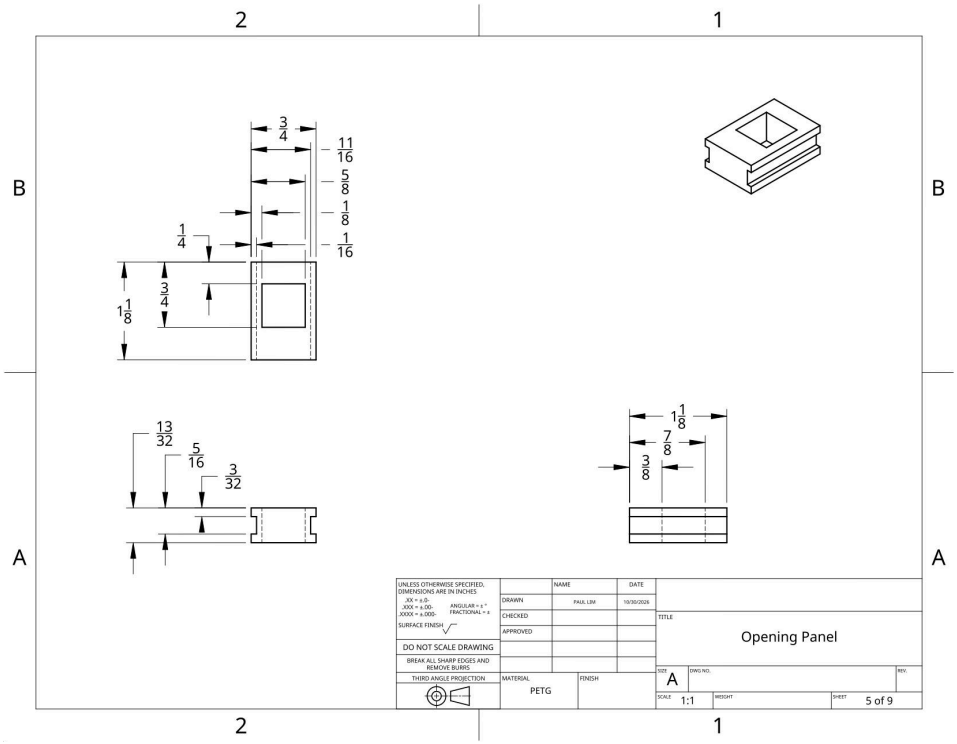


Figure 5. Dimensions and sketches of prototype's opening panel, allowing a syringe to be inserted and attached to the prototype

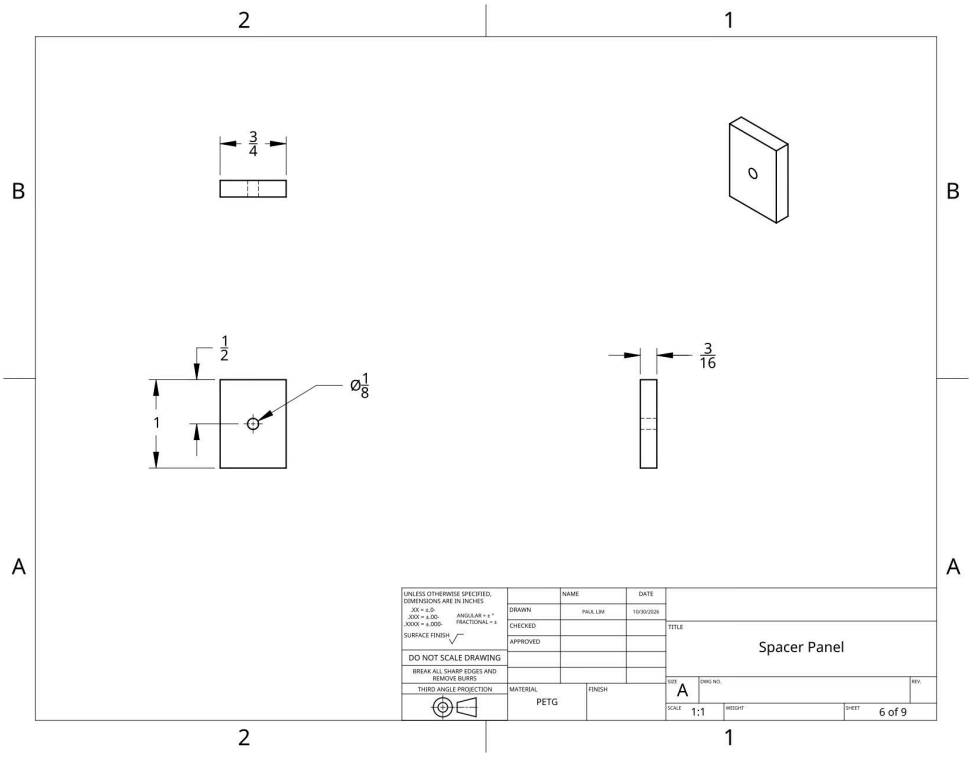


Figure 6. Dimensions and sketches of panels meant to help space components in the prototype

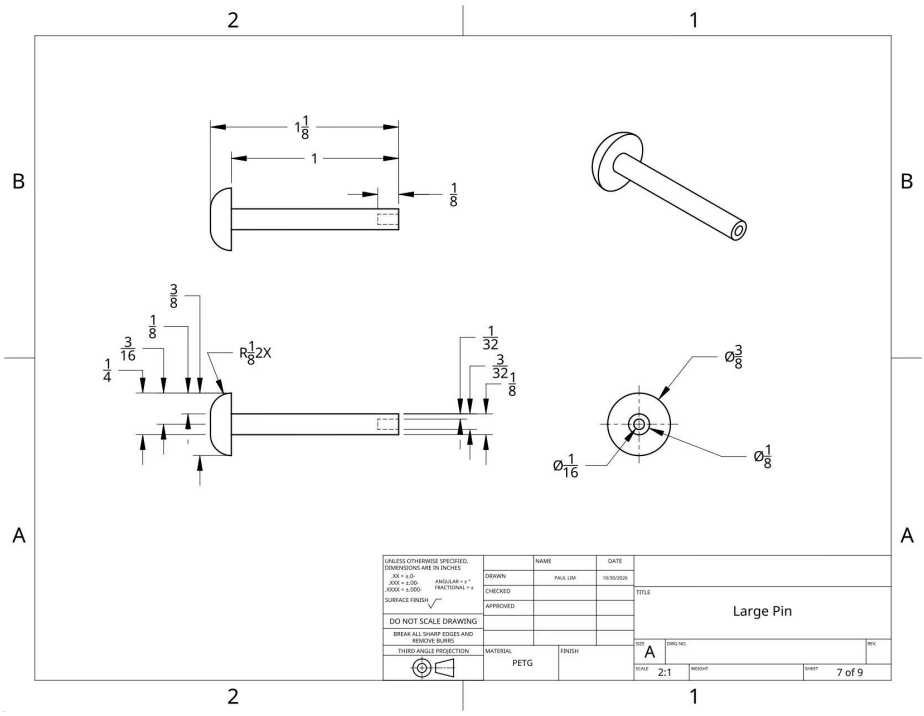


Figure 7. Dimensions and sketches of a large pin that hold the 2L bar and L bar in place with the prototype's casing

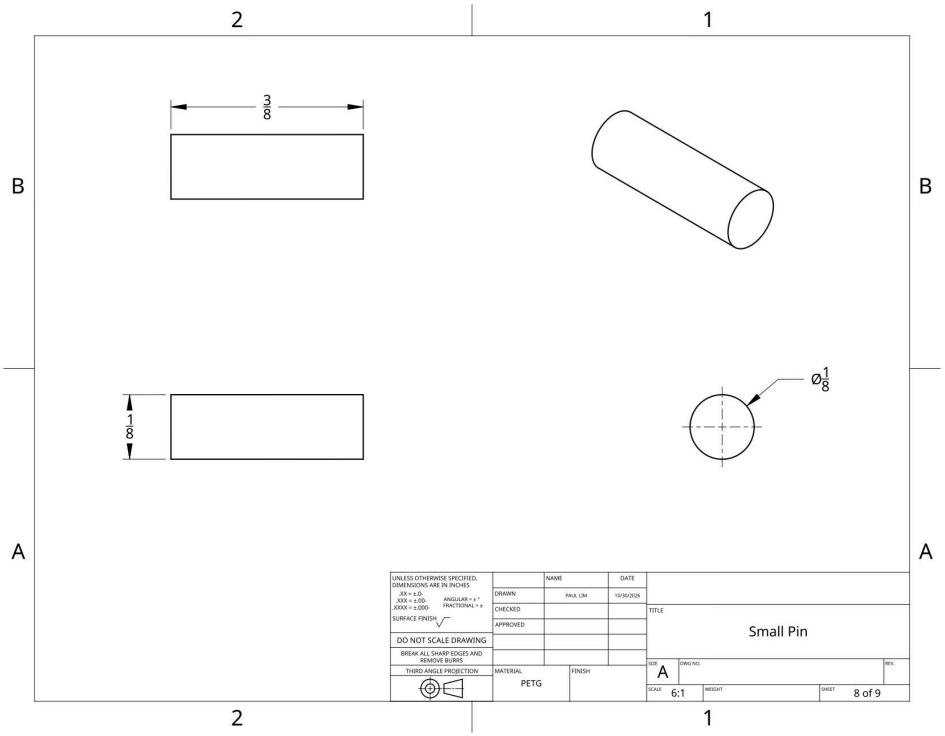


Figure 8. Dimensions and sketches of a pin that would hold the 2L bar and L bar together at the midpoint of the 2L bar.

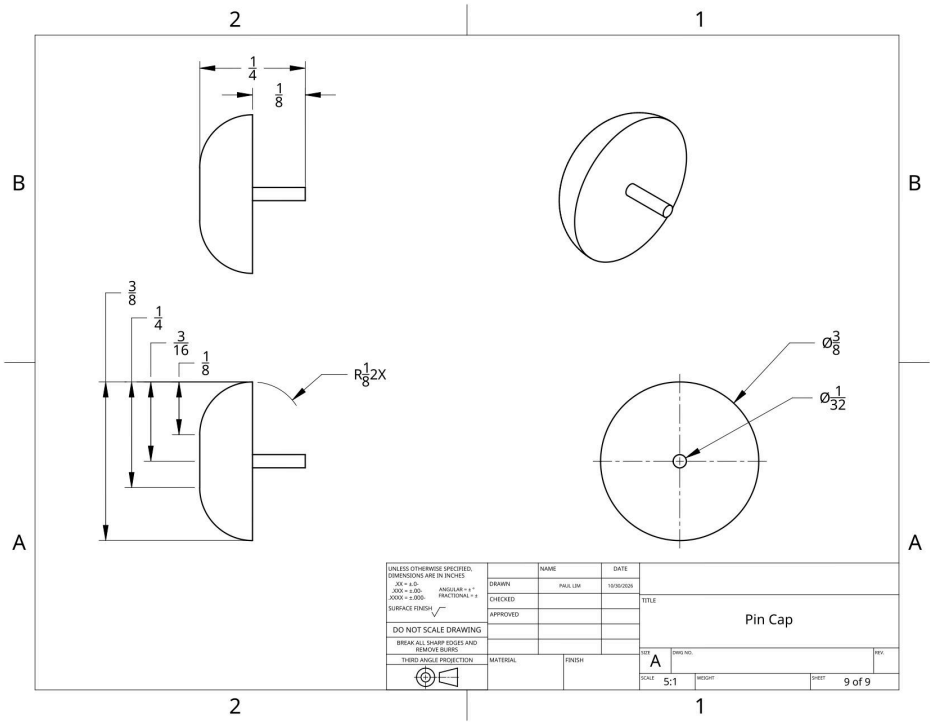


Figure 9. Dimensions and sketches of a cap to the large pins to prevent the pins from slipping out of the casing and other components during operation.

6.4. **Functional Block/Subcomponent Diagram or Flowchart**

Table 1 - Subcomponent Table

Subcomponent Title	Technical Description	Contributing discipline/ skillset	Design Specification Addressed	Team Member Assigned
Existing Commercial Syringe	Standard commercial 1 mL Luer-lock syringe with 32 gauge needle (ex. BD 2189 [3]) supplied as-is.	Clinical / healthcare / medical	Must be compatible with off-the-shelf 1 mL syringes (1 mL barrel, 32G needle)	Samika
Fixed Frame Mount	Rigid bracket that anchors the lever's rotation axis to the device housing and clamps onto the syringe barrel. Fixes the mechanism's geometry and holding the syringe/needle steady to a target user-rated stability of >⅘ (ideal 5/5, marginal ⅗)	Mechanical engineering	Must improve syringe/needle stability (user-rated >⅘, ideal 5/5, marginal ⅗)	Samika
Driving Lever Arm (Thumb-Actuated Input)	Handle sized for a natural one-handed grip, so the full injection is actuated and held steady with one hand, targeting a comfort rating of >⅘ (ideal 5/5, marginal ⅕)	Mechanical engineering / human factors	Must be usable one-handedly and prioritize patient comfort (customer rating >⅘, ideal 5/5, marginal ⅕)	Paul
Scott Russell Coupler Link	Equal-arm rigid linkage that converts physician-applied input force into axial plunger-driving force through rotational equilibrium and linkage kinematics. Mechanical advantage varies with the linkage angle, with POC calculations indicating effective force amplification at < 37.85 degrees under the tested loading conditions. Controlled force also aims to minimize excessive plunger pressure and associated silicone oil droplet release into the injected medication.	Mechanical engineering	Must reduce injection force (<0.40 N ideal / <0.44 N marginal delivery force) [4]. Must not increase silicone oil particulates (< USP<789> limits).	Paul
Output Slider (Plunger-Drive Carriage)	Carriage that contacts the syringe's plunger thumb-press directly, so the clinician	Mechanical engineering	Must not make procedure more difficult (hand force	Erica

	depresses the plunger through the device's single actuation motion rather than a separate manual push, holding the hand force required at the slider to <1.0 N (ideal) / <3.0 N (marginal) [5].		<1.0 N ideal / <3.0 N marginal)	
Linear Guide Rail	Fixed linear guide that constrains the output slider to one translational degree of freedom along the syringe axis, minimizing lateral plunger displacement and maintaining alignment during medication delivery. Slider travel is mechanically limited to the prescribed dose, with a target injection rate of 0.01-0.05 mL/min for a 32G needle [6].	Mechanical engineering / fluid mechanics	Must be able to support an injection rate of 0.01 mL/min with a 32G needle (ideal 0.6 mL/min / marginal 0.01 mL/min)	Erica
Sterilizable Housing	Sterilizable housing encloses the linkage without contacting the fluid pathway and supports validated sterilization with SAL < 10^{-6} [7].	Biomedical engineering / materials science	Must be sterilizable (SAL < 10^-6)	Vishnu
Tapered Needle	Internal funnel taper (8 mm inlet narrowing to a 3 mm bore over a 4 mm cone) fitted ahead of the standard 32 gauge lumen; CFD modeling of the modified geometry showed a 31.5% increase in driving pressure needed to match the unmodified needle's flow at the 0.01-0.5 mL/min target range [8].	Mechanical engineering / fluid mechanics	Must be able to support an injection rate of 0.01 mL/min with a 32G needle.	Jasmine

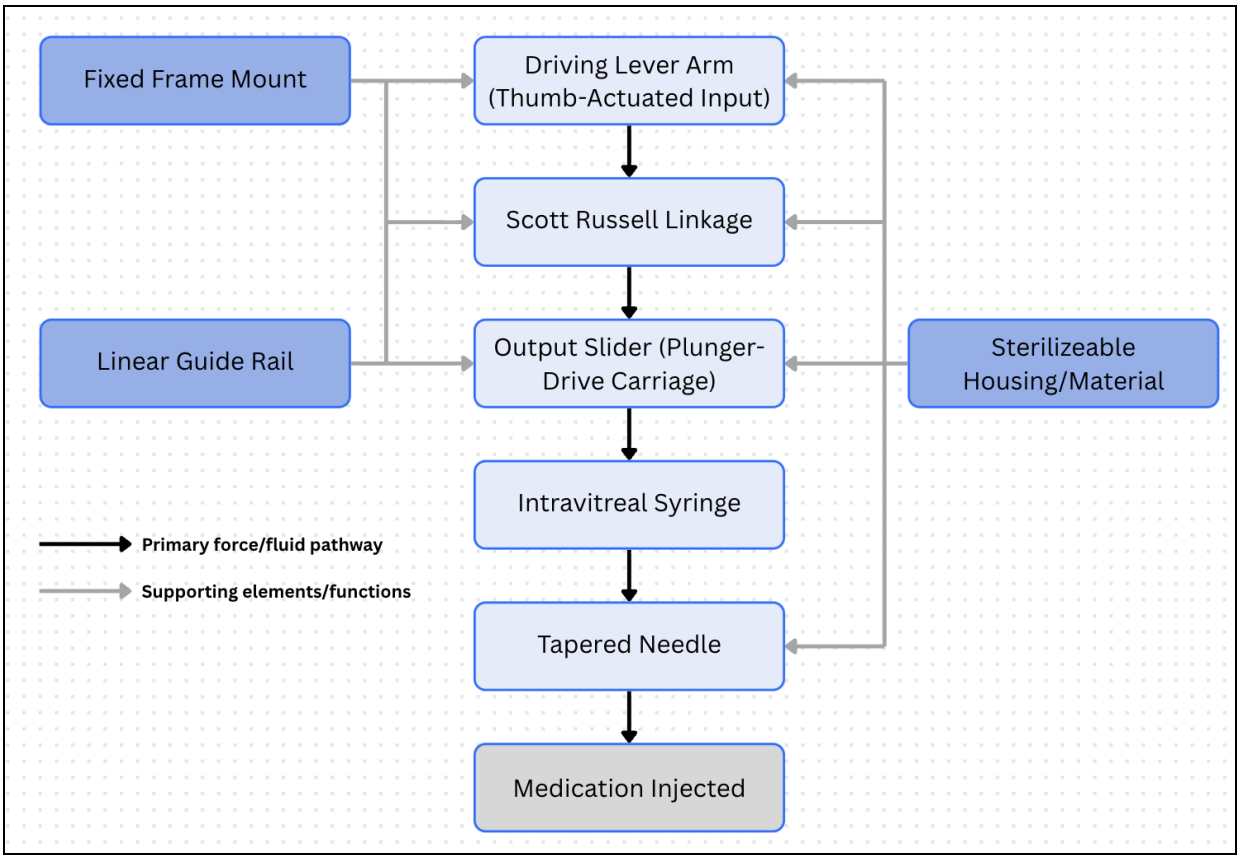


Figure 10 - Subcomponent flowchart

References

[1] “The effect of syringe design and cannula dimensions on time-force curve in intravitreal injection across different drug viscosities: Area under the curve and peak injection force,” PubMed Central. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC13063457/

[2] “Transient vision loss associated with prefilled aflibercept syringes: A case series and analysis of injection force,” PubMed Central. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9541561/

[3] Becton, Dickinson and Company, “BD 2189 Disposable Single-Use Syringes — Technical Data Sheet.” [Online]. Available: https://www.bd.com/content/dam/bd-assets/na/medication-delivery-solutions/documents/materials-safety-data-sheet/bd-2189-disposable-single-use-syringes-technical-data-sheet.pdf. [Accessed: Sep. 18, 2026].

[4] L. Christensen, A. Cerda, and J. L. Olson, “Real-time measurement of needle forces and acute pressure changes during intravitreal injections,” Clinical & Experimental Ophthalmology, vol. 45, no. 8, pp. 820–827, Nov. 2017, doi: 10.1111/ceo.12963. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/28418609/

[5] J. J. Raevis, C. A. Lemire, D. J. Ramsey, J. Riccobono, and E. Gonzalez, “Deformation of aflibercept and ranibizumab syringes causes variation in intravitreal injection volume and risks retinal tear formation,” Ophthalmology Science, vol. 2, no. 4, art. 100202, 2022, doi: 10.1016/j.xops.2022.100202. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9754977/. [Accessed: Sep. 18, 2026].

[6] H. Lee, Y. Lee, J.-Y. Park, and J. Han, “Analysis of flow rate and pressure in syringe-based wound irrigation using Bernoulli’s equation,” Scientific Reports, vol. 12, art. no. 14957, Sep. 2022, doi: 10.1038/s41598-022-19402-2. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9440234/

[7] StaClear, Inc., “StaClear — The First Low-Particulate Intravitreal Syringe.” [Online]. Available: https://www.staclear.com/. [Accessed: Sep. 18, 2026].

[8] E. Krayukhina, A. Fukuhara, and S. Uchiyama, “Assessment of the injection performance of a tapered needle for use in prefilled biopharmaceutical products,” Journal of Pharmaceutical Sciences, vol. 109, no. 1, pp. 515–523, Jan. 2020, doi: 10.1016/j.xphs.2019.10.033. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0022354919306707